MAE 3 Robot Report (Team 27)

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Description of Component

"Nobody Has Ever Seen Such Spectacular Engineering" (N.E.S.S.I.E.) is a remote-controlled stationary robot consisting of one hook arm that was designed for the Spring 2025 MAE 3 Robot Competition. The primary objective of the robot is to collect rings scattered around one side of the table within one minute using its multiple-jointed arm. The robot would pick up the rings using a hook attached to the end of the arm and place them on the Sun God statue. The robot has four main components: the base arm, the lower arm, the upper arm, and the hook.

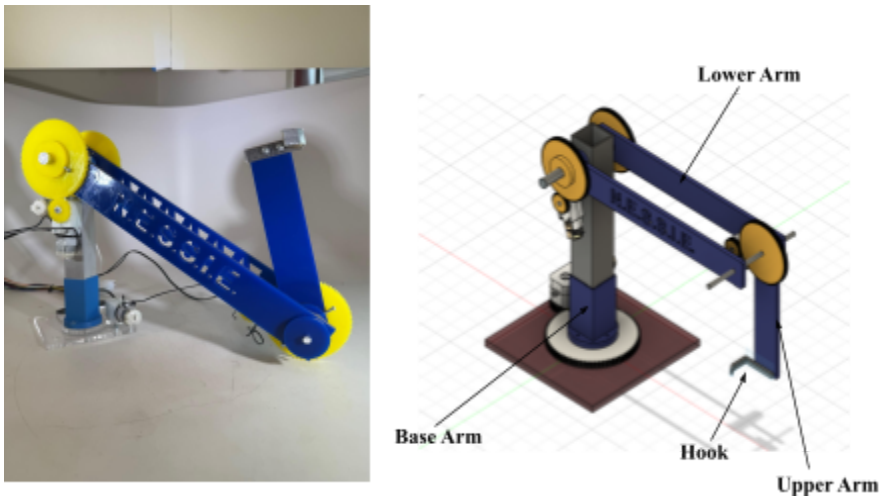


Fig 1. Picture of the robot N.E.S.S.I.E. in its initial position (left), CAD of final N.E.S.S.I.E. robot design (right).

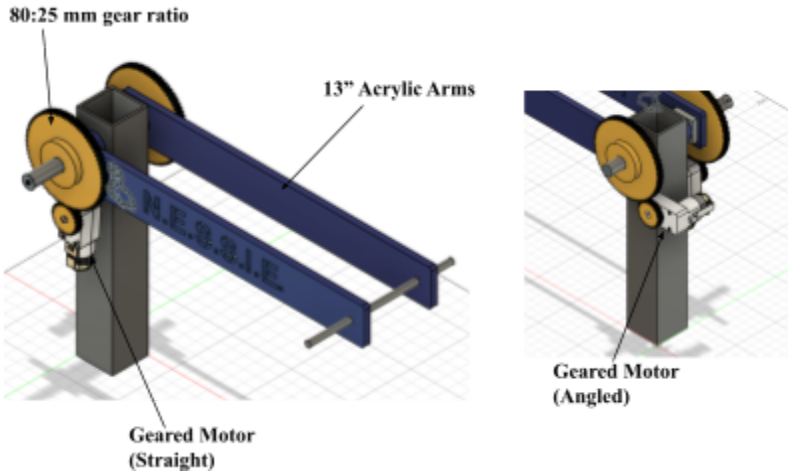


Fig 2. Isometric View of Lower Arm Component for N.E.S.S.I.E. (left), isometric back view of N.E.S.S.I.E. showing angled gear motor (right).

The design of the robot is rather simple: the base arm rotates the arm around a swivel, while the two arms can move up and down using simple gear ratios, with the upper arm containing the hook to grab rings.. For this report, the lower arm component of the robot (the component connected to the base arm and upper arm) will be analyzed. For the N.E.S.S.I.E. design to work, the lower arm component must meet these functional requirements (FR):

- The arm must hold the weight of the upper arm and the hook.
- The Lower Arm must be able to rotate 45° or -45° from its initial position (horizontal) in 2 seconds.

This report will analyze the first FR. To test the maximum load, a weight of arbitrary magnitude was placed at the end of the hook. The motors would then lift the arm to see if the arm can handle that weight. This test was repeated with heavier weights until the arm began to stall and the motors were no longer able to move the arm past 0° (perpendicular to the base arm). From initial calculations before the start of the experiment, it was calculated that the arm could only lift 35 g before the motors attached to the lower arm would stall. However, it was recorded that the maximum weight the arm could hold was ~140 g. This value reflects how much load the lower arm can withstand at the end of the hook; this load does not include the weight of the arm components themselves. While the theoretical and experimental values differ tremendously, the values do make sense with the given assumptions made in the analysis (explained further in the Analysis section). While the theoretical values make sense under the assumption that both motors output the same amount of torque, it was found that the motors output a different amount of torque, with the L-shaped geared motor having a greater maximum torque.

Analysis of Component

The focus of this analysis is to answer this question: "What is the maximum amount of mass the lower arm can handle?" To answer this question, the F.S.$_{torque}$ must be determined to find whether or not the two geared motors can hold the weight of the lower arm. For this analysis, the following assumptions were made:

- The arm system is in quasi-static motion.
- The friction between the motors is negligible.
- The weight of the arms is assumed to occur at the center of mass (COM), and the hook will be treated as a point mass.
- The input and output gear weights are negligible.

Using these assumptions, the following Free Body Diagrams were drawn:

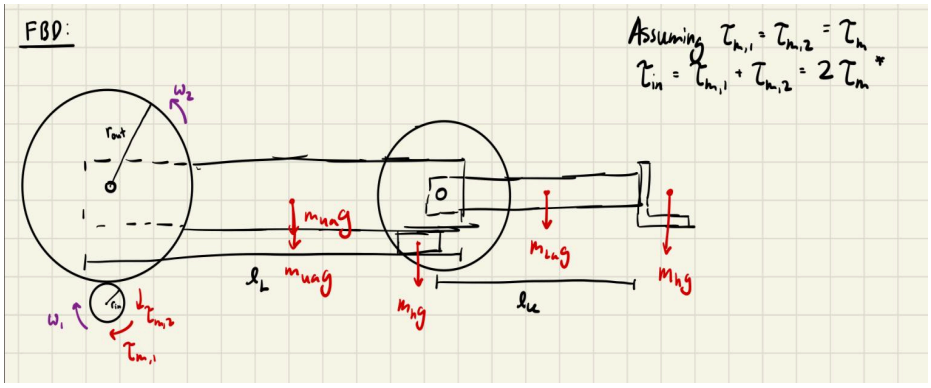
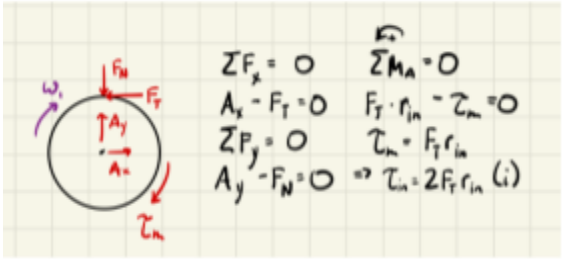


Fig 3. Free Body Diagram of Arm System



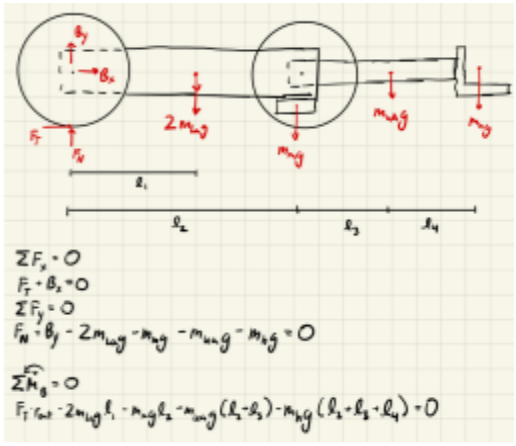


Fig 4. Free Body Diagrams and Quasi-Static Equations for the Input Gears (left) and the Lower Arm Component (right), with forces and lengths labeled for each diagram.

From Fig. 4, multiple distances were measured to reduce unnecessary calculations and for accuracy. Below is a description of each length.

Table 1. Length Number and Description of What They Measure

Length #	**Description**
1	Distance between the output gear center and the center of mass (COM) of the lower arm.
2	The distance between the output gear center and the center of the hole where the upper arm and lower arm connect.
3	Distance between the hole where the lower and upper arm connect and the COM of the upper arm.
4	Distance between the COM of the upper arm and the hook

From these FBDs and the initial assumptions made for the analysis, the main equations that are crucial in solving the questions are:

$$\Sigma M_A = 0 \qquad (1)$$

$$\Rightarrow F_T = \frac{\tau_{needed}}{2 \cdot r_{in}} (2)$$

$$\Sigma M_B = 0 \qquad (3)$$

$$\Rightarrow \tau_{needed} = \frac{2 \cdot r_{in} g[2 \cdot m_{la} l_1 + m_m l_2 + m_{ua}(l_2 + l_3) + m_h(l_2 + l_3 + l_4)]}{r_{out}} \quad (4)$$

Eqs. (1) and (3) are a result of the quasi-static motion assumption and finding the moments about the center point of the input gears (A) and the output gear attached to the lower arm (B). Eqs. (2) and (4) aim to solve crucial variables such as the tangential force on the input gear from the

output gear $F_{T,}$ and the torque needed to keep the arm parallel to the ground (0°). Eqn. (5) is needed to find the factor of safety concerning the torque:

$$F.S._{\tau} = \frac{\tau_{in}}{\tau_{needed}} (5)$$

where τ_{in} is the actual output torque provided by the two geared motors.

To obtain the values for a theoretical analysis of the torque, the parts were analyzed using Fusion 360 CAD software by checking the properties of the part and utilizing the "Measure" tool to determine the lengths. Tables 2 and 3 show the values generated from Fusion360:

Table 2. Theoretical Values for Mass of Components

Component	**Mass (g)**
Lower Arm (m_{la})	84.551
Upper Arm (m_{ua})	54.512
Motor (m_m)	26.0
Hook (m_h)	8.0

Table 3. Theoretical Values for Length of Components

Component	**Length**
Input Gear Radius (r_{in})	12.5 mm
Output Gear Radius (r_{out})	40 mm
Length 1 (l_1)	14.732 cm
Length 2 (l_2)	29.718 cm
Length 3 (l_3)	7.874 cm
Length 4 (l_4)	17.399 cm

Using these values obtained from Fusion360, Eqs. (2) and (4) result in:

$$\Rightarrow F_T = 14.11\,N \qquad (6)$$

$$\Rightarrow \tau_{needed} = .353\,Nm \qquad (7)$$

From the Force-Torque-Power-Energy (FTPE) analysis conducted before the robot's fabrication, the geared motors were found to have a maximum torque (τ_m) of 0.271 Nm. See Appendix A for the full analysis of the powered gear components.

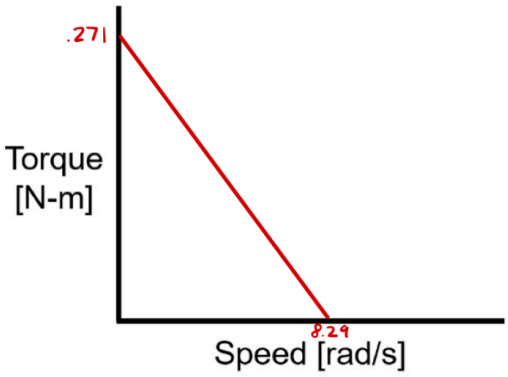


Fig 5. Graph of the Torque (Nm) versus the angular speed (rad/s) for the Geared Motor

Knowing that the lower arm is powered by two geared motors, the F.S. for the torque can be calculated:

$$F.S._\tau = \frac{\tau_{in}}{\tau_{needed}} = \frac{2{\cdot}.271}{.353} = 1.54 \qquad (8)$$

With a factor of safety of 1.13, this states that the torque provided by the two geared motors is sufficient to lift and sustain the weight of the arm. However, to find how much additional weight the arm can sustain, the following calculations are done:

$$\tau_{additional} = \tau_{in} - \tau_{needed} = .189\,Nm \qquad (9)$$

$$\tau_{additional} = W \cdot r = m_{additional} g(l_2 + l_3 + l_4) \qquad (10)$$

$$\Rightarrow m_{additional} = \frac{\tau_{additional}}{g(l_2+l_3+l_4)} = .035\,kg \Rightarrow 35\,g \qquad (11)$$

Thus, the theoretical maximum weight the motors can handle, assuming that additional weight is placed on the hook of the arm, is ~35 g.

To find the experimental max weight of the robot, weight was added to the end of the hook until the motors stalled the arm (could not lift the arm any higher than its initial position). From the experiment, a weight of ~140 g was added to the arm before the motors began to stall. From error analysis, it was identified that the percent error of the additional max weight was:

$$m_{\%\ error} = \frac{|m_{add,calc} - m_{add,act}|}{m_{add,act}} \cdot 100 = 75\% \quad (12)$$

Discussion

From the results of the analysis, it was observed that the experimental value of the maximum mass lifted was significantly higher than what was calculated. Many discrepancies could be attributed to why a high percentage error of 75.0 % was recorded. To start, the arm component system is much more complex in reality compared to what was drawn from the FBDs. The FBDs assumed that all additional components in the lower arm (i.e. steel rods, spacers, screws) are all treated as sharing the same COM as the acrylic parallel arms. This would mean that not all weight was concentrated at the center of each arm, and thus would result in a much lower torque needed to lift the arm. The arms and all bodies also had irregular shapes and characteristics (i.e., holes, triangular patterns on the side) that would not make a point-mass analysis an accurate way of analyzing the maximum weight that can be carried. Additionally, one important to note is that the geared motors used for the FTPE analysis were inconsistent; while the straight and angled geared motor are theoretically valued to provide the same amount of torque, further testing concluded that the angled geared motor provided more torque than the straight geared motor, thus increasing the actual $F.S._{\cdot torque\cdot}$

The chosen two-jointed limb stationary robot was advantageous for the goal of this contest, however, there were many limitations associated with the design. While the arm joints had an incredible reach and did not require any complex maneuvers to obtain rings and place them on the Sun God statue, the design struggled in obtaining rings that were too close or just out of its reach. The placement of N.E.S.S.I.E. also limited its point potential for the competition, as the potential for scoring points and reaching the Sun God statue for possible point multipliers heavily depended on the starting point of N.E.S.S.I.E.

If given a second chance to create and manufacture another robot for this current competition, I would heavily consider using a 4-bar linkage design for the arm rather than a simple two-bar design. The reason for this change is that a 4-bar linkage would provide more stability for the arm mechanism, and robot maneuvering would be smoother, as it was noted that it was harder to maneuver the robot due to the weight being an issue in lifting.

From this project, the biggest technical lesson that I took away is how crucial it is to go through multiple iterations and prototypes to test a multitude of different designs. Looking back, I believe that the N.E.S.S.I.E. robot was deemed successful, as the main focus for the robot was to score as many points as possible by reducing the complexity of obtaining points. With a simple crane design, N.E.S.S.I.E. exceeded the team's expectations for the competition, as it scored high despite the lack of translational movement throughout the table. The design, however, could have increased its potential in scoring points if other designs for the arm were considered (i.e., 4-bar linkage, rotating hook).

Design Process

Initially, in the early stages of design for N.E.S.S.I.E., the idea of having a stationary robot that would have a long enough arm to reach all aspects of the table seemed doable and efficient for the competition. The long arm would compensate for the lack of a drivetrain, thus removing the need to freely move around the table. When designing ideas of what could be done for the risk reduction, the group was fixated on whether the geared motors would provide enough torque and speed to rotate and support the arm in its riskiest position (parallel with the ground).

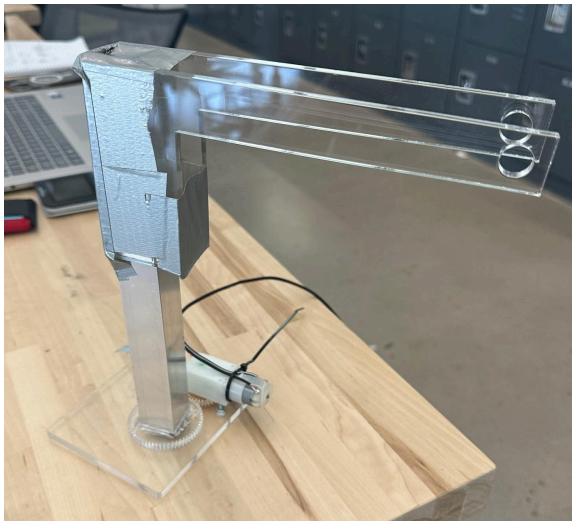


Fig. Risk Reduction for N.E.S.S.I.E. base arm component.

The idea for the risk reduction was to develop a rotating arm that would freely revolve about the base plate, and to test how the amount of weight can be held at the end of the arm without the speed decreasing, and to keep the arm stable at all times during the rotation. From the numerous risk reduction tests, the group concluded that the biggest problem was the output gear lifting off the ground, and that the steel arm would need to be supported to keep it stable. The solution the group came up with was to have a 3D-printed holder that would hold the steel arm in place, and a hubcap that would be placed inside the holder to keep the gear and the holder from lifting off the ground during rotation.

Appendix

A. Force-Torque-Power-Energy Analysis for Given Powered Components

Question 3: Team Force Torque Power Energy Assignment (300pt)

Team Number: 27 Team Members: Joshua Conge, Pranav Ramesh, Isaac Kim, Connor Lin

At the beginning of a machine design project, it is important to quantify your energy sources. Each quarter, the kit varies, and each team must measure Force, Torque, Power, and Energy in their kit.

Assignment Details: Each team should complete this worksheet by measuring values in the lab, performing the analysis, and then filling the values into the tables below (tolerance of ±20%). Each individual will submit their copy of this assignment to Gradescope with all boxes and plots filled in, but all work should be shared between the team. In addition, Individual components are to be split up among team members for their energy analysis (Question 1). Show details of work neatly.

Potential Energy from Gravity:

Determine the maximum amount of energy available due to gravitational potential. Assume that half the total mass of the kit is free-falling from the maximum contest start height.

Mass - ½ of kit (kg):	m_{max}	2.89 kg
Height Change (m):	h_{max}	.127 m
Release Time free-fall (s):	$t = \sqrt{2h/g}$	.161 s
Potential Energy (J):	$E = mgh$	3.60 J
Power (watts):	$P = \frac{E}{t}$	22.36 W

Springs and Rubber Bands:

Experimentally determine the spring constant ‘k’, then calculate the energy and power available from the springs/rubber bands. Make sure masses are large enough to overcome the initial spring pretension. (The springs labeled as Small diameter and Short (SS), Small diameter and Long (SL), and Large diameter and Long (LL) *max elongation is 3 times the length of the rubber band*

		Small RB	**Med RB*** **	**Large RB**	**SS SP**	**SL SP*** **	**LL SP**
Load 1 (N):	F_1	2.8	5.8	2.8	2.8	2.117	2.8
Length (m):	L_1	0.111	0.218	.115	.013	0.044	.038
Load 2 (N):	F_2	4.1	2	4.1	4.1	2.42	4.1

Length (m):	L_2	0.142	0.127	.143	.018	0.051	.052
K (N/m):	$k = \frac{F_2 - F_1}{L_2 - L_1}$	41.94	41.7	46.43	260	43.28	92.85
Initial Length (m):	L_o	0.0927	0.101	.0889	.009	0.035	.0285
Max Length* (m):	L_m	0.278	0.303	.266	.0508	0.139	.177
Max Elongation (m):	$\delta_x = L_m - L_o$	0.1855	0.202	.1777	.0418	0.1047	.149
Max Force (Nm):	$F = k \cdot \delta_x$	7.779	8.3	8.250	10.868	4.53	13.85
Max Energy (J):	$E = \frac{1}{2} k \cdot \delta_x^2$	0.7215	0.82	0.733	0.227	0.237	1.033
$T_{release}$ (s):**	$t_{release}$	0.156	0.16	0.149	0.063	0.05	0.105
Max Power (watts):	$P = \frac{E}{t}$	4.599	5.12	4.916	3.604	4.74	9.804

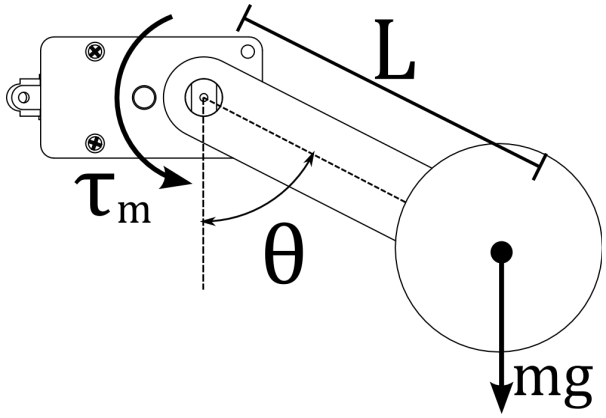
*** Max Length:** Do not damage springs by stretching them beyond their elastic limit., We have pre-stretched springs in the design studio for you!

**** Release Time: A** good estimate for the release time is ¼ period of oscillation. Estimate the mass the spring will be moving and calculate the natural period $T = 2\pi\sqrt{m/k}$.

*** Values provided by Tutors.

DC Motors:

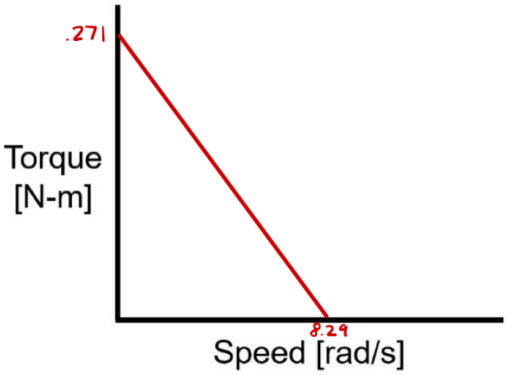
Using the setup in the lab, measure and calculate the torque, speed, and power of the geared and non-gearred motors. Using your results, plot the torque-speed motor curves.



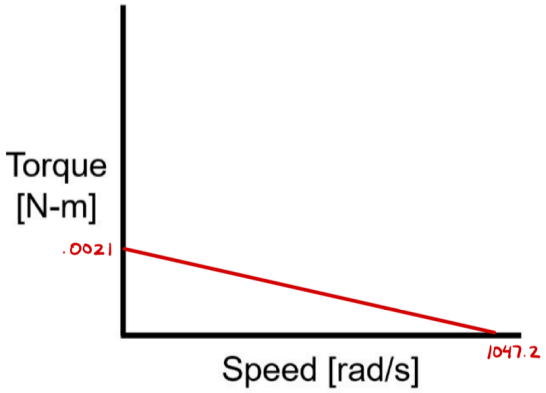
		Geared Motor	Non-Geared Motor
Stall Angle (deg):	θ	47	19
Moment Arm Length (m):	L	.13	.065

Mass (kg):	m	.291	.01
Stall Torque (Nm):	$\tau_{stall} = mgLsin\theta$	.271	.0021
No Load Speed (rad/s):	ω_{noLoad}	8.29	1047.2
Max Power (watts):	$P = \frac{1}{2}\tau_{stall} * \frac{1}{2}\omega_{noLoad}$	.561	.550
Max Energy in 60s (J):	$E = P \cdot t$	33.7	33

Geared Motor:



Non-Geared Motor:



Summary of Energy Sources:

	Max Force/Torque:	**Max Energy:**	**Max Power:**
Geared Motor	.271 Nm	33.7 J	.561 W
Non-Geared Motor	.0021 Nm	33 J	.550 W
Small Rubber Band	7.779 N	0.721 J	4.599 W
Med Rubber Band	8.3 N	.82 J	5.12 W
Large Rubber Band	8.250 N	0.733 J	4.916 W
Small Spring	10.868 N	0.227 J	3.604 W
Medium Spring	4.53 N	.237 J	4.74 W
Large Spring	13.85 N	1.0337 J	9.804 W
Weight (Pot. Energy)	28.35 N	3.60 J	22.36 W

	Component:	Value:
Max Force:	Large Spring	13.85 N
Max Torque:	Geared Motor	.271 Nm
Max Power:	Weight (Pot. E)	22.36 W
Max Energy:	Geared Motor	33.7 J